IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – [Write your project title here]

Domain of Project – [like Agriculture / Education / Healthcare / …]

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Problem Statement – Research Agent

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The Challenge
Researchers, students, and professionals often deal with scattered academic papers, technical articles, and experimental data spread across multiple platforms. Filtering relevant knowledge, summarizing insights, and keeping track of the latest developments is time-consuming and inefficient. The challenge is not just to collect references but to synthesize, analyze, and generate actionable research insights in real time.

The Objective
Build an agentic AI application that acts as an intelligent research companion. The solution should:
1. Fuse Academic Sources: Aggregate data from research papers, journals, preprints, and knowledge repositories, then synthesize them into clear, concise literature reviews.
2. Enable Multimodal Research Input: Accept user queries, notes, charts, or even PDFs, and convert them into structured insights, references, and summaries.
3. Create a Predictive & Agentic Layer: Highlight emerging trends, detect citation gaps, and predict promising directions for future research in a given domain.
4. Visualize Research Insights: Provide an interactive dashboard with topic clusters, citation maps, sentiment/trend analysis, and concise knowledge graphs

Technology Used

Tech Stack (Mandatory)
Use **any one** of the following approaches:

- LangFlow + IBM watsonx.ai + Granite Models
- IBM Cloud Agentic Lab with IBM Granite Models
- Replicate with LangChain + RAG + IBM Granite Models

I used LangFlow + IBM watsonx.ai + Granite Models

Proposed Solution – Research Agent

Proposed Solution: Research Agent
The proposed solution is a multi-agent AI-powered Research Agent built using LangFlow, IBM watsonx.ai, and IBM Granite Models to assist students, researchers, academicians, and professionals in efficiently conducting research, analyzing documents, and generating intelligent insights. The system addresses challenges in literature review, research data organization, information overload, and academic analysis through an intelligent modular architecture.

At its core, the solution uses a Retrieval-Augmented Generation (RAG) pipeline to retrieve relevant information from research papers, journals, PDFs, online repositories, and academic databases. The collected data is processed into embeddings and stored in a vector database, enabling semantic search and context-aware retrieval of research content. IBM Granite models are then used to summarize, analyze, and generate meaningful responses from the retrieved information.

The system consists of four key agents:
Research Knowledge Agent retrieves and summarizes research papers, articles, and academic documents using RAG-based semantic retrieval. It extracts important concepts, methodologies, conclusions, and research gaps from uploaded or online sources.
Literature Review Agent generates structured literature reviews by combining insights from multiple research papers. It compares methodologies, highlights trends, identifies similarities and differences, and organizes information into a research-friendly format.
Insight & Analysis Agent performs intelligent analysis on collected research data by generating summaries, extracting keywords, identifying patterns, and providing contextual answers to user queries. It helps researchers quickly understand complex information and make informed decisions.
Citation & Report Generation Agent assists users in generating citations, references, and research reports in structured formats. It supports academic writing by organizing findings, creating concise summaries, and improving documentation efficiency.

Langflow component Used -

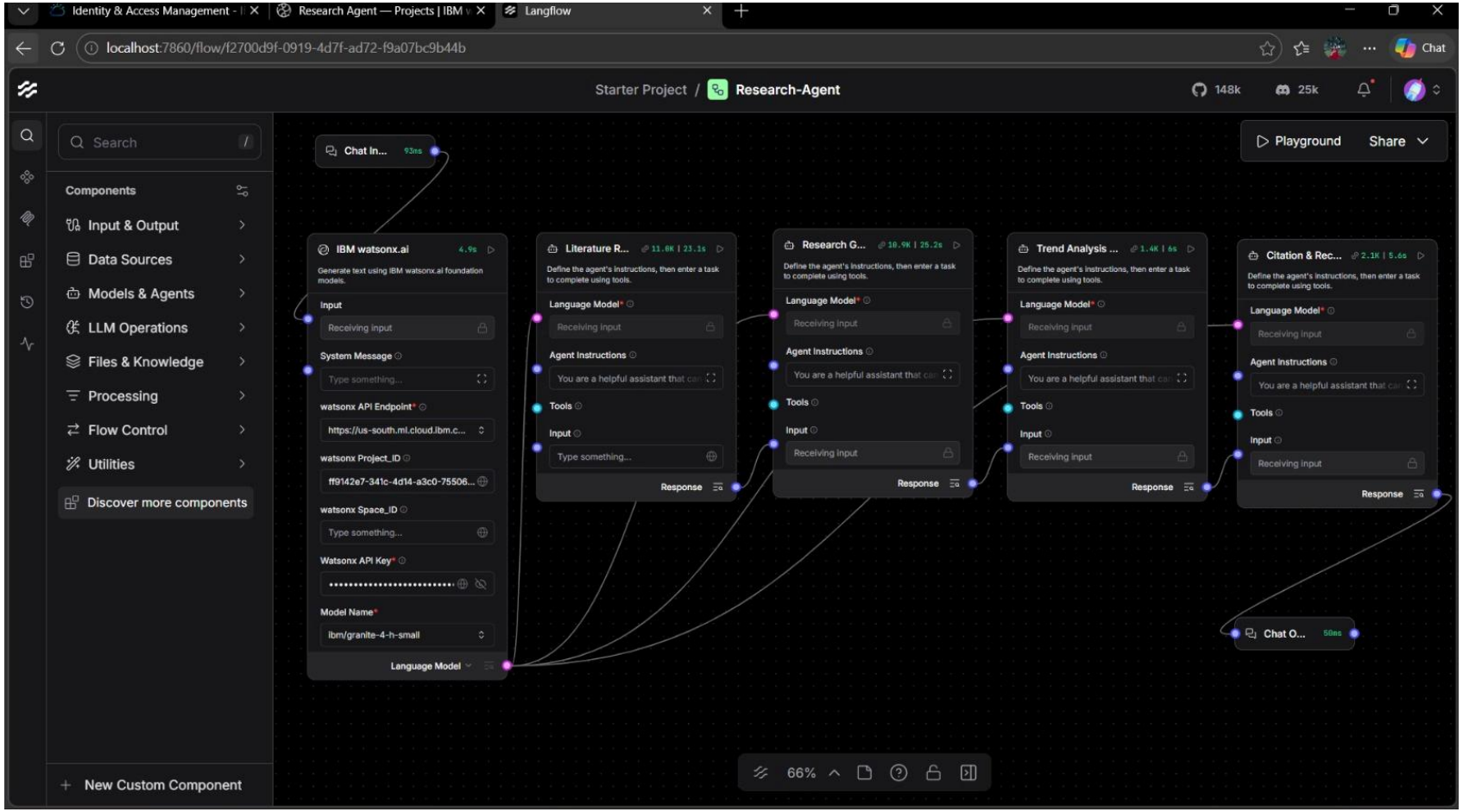
1. **Literature Review Agent** – Retrieves, summarizes, and compares research papers to generate structured literature reviews.

2. Research Gap Agent – Identifies unexplored areas, limitations, and future research opportunities from existing studies.

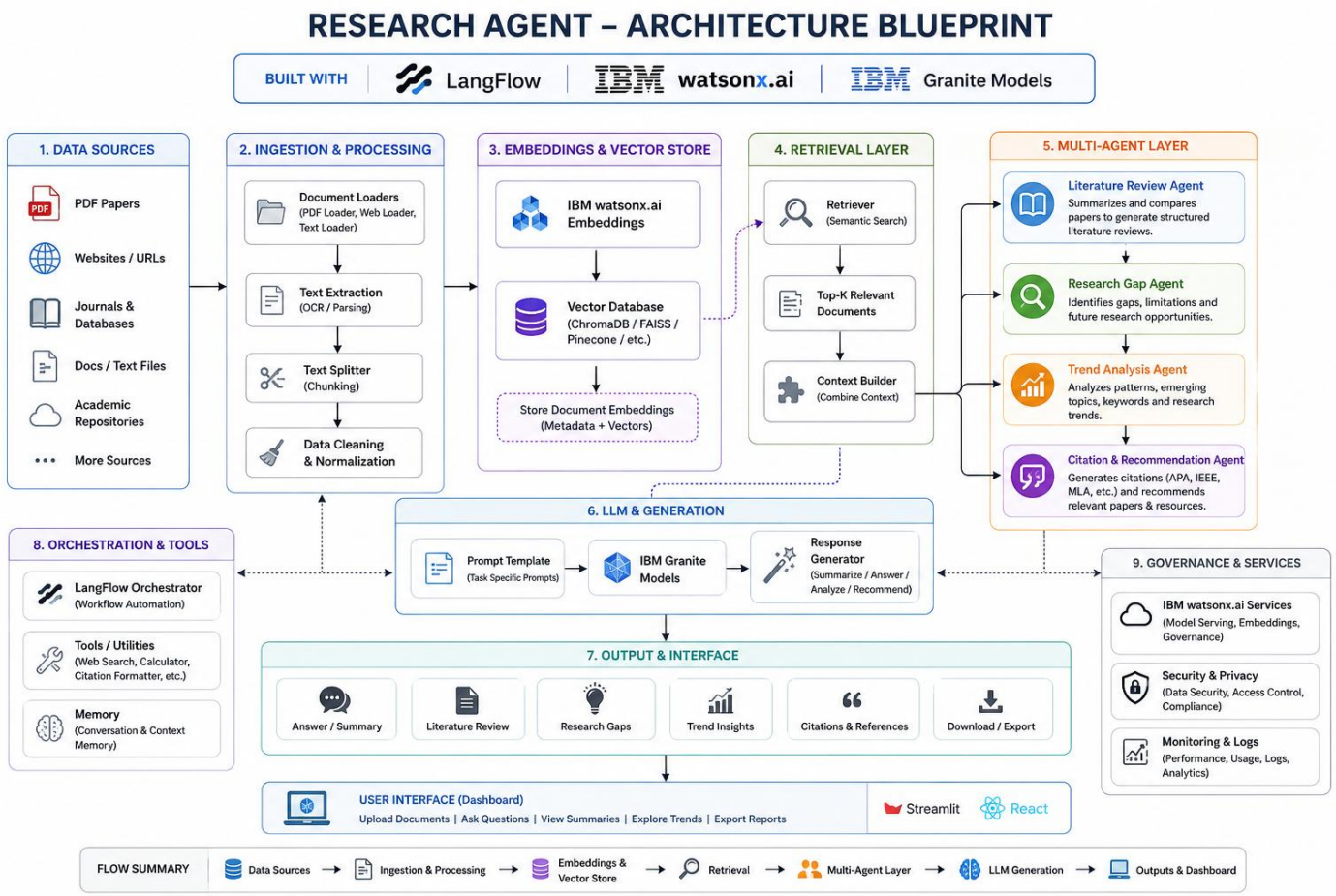
3. Trend Analysis Agent – Analyzes research patterns, emerging technologies, and trending topics across domains.

4. Citation & Recommendation Agent – Generates accurate citations and recommends relevant research papers and resources.

Langflow workflow -



Architecture blueprint



Role of Agentic AI in the solution

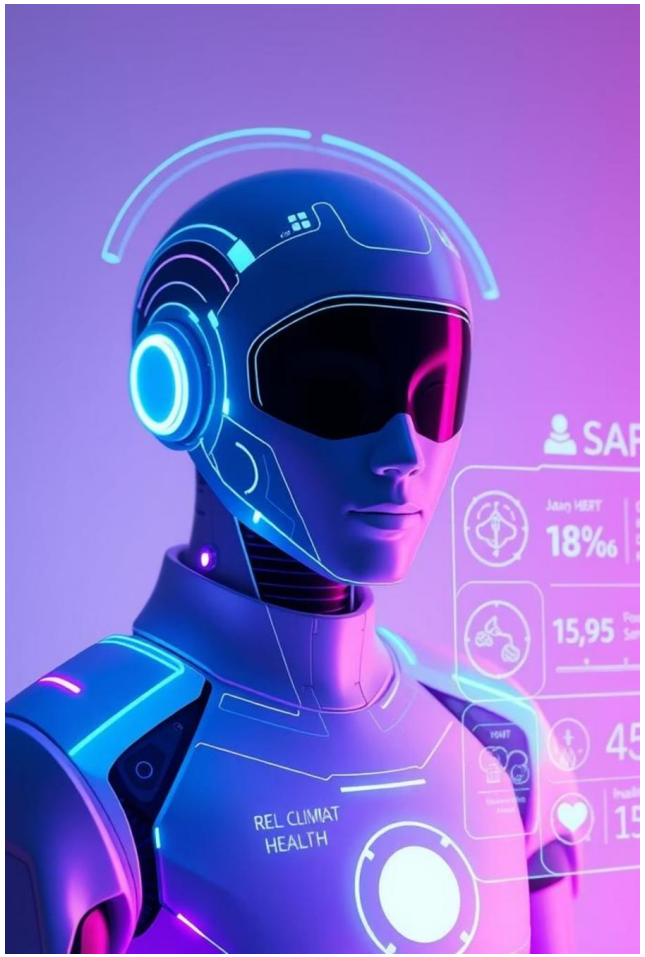
Role of Agentic AI in Research Agent

Agentic AI enables the Research Agent to function as an intelligent, autonomous research assistant rather than a basic question-answering system. Using IBM watsonx.ai and IBM Granite models, the system coordinates multiple specialized agents to perform complex research-related tasks efficiently and collaboratively.

Each agent is designed to handle a specific function—such as generating literature reviews, identifying research gaps, analyzing trends, and managing citations—while working together to provide accurate and context-aware research insights. This multi-agent architecture improves task distribution, enhances efficiency, and delivers high-quality research assistance.

Agentic AI also provides contextual understanding by analyzing user queries, uploaded research papers, academic domains, and research objectives to generate personalized and relevant outputs. It supports real-time adaptability by continuously updating responses and recommendations based on newly uploaded documents, changing research topics, or user interactions.

Additionally, the agents communicate and share retrieved information with each other, enabling intelligent decision-making and deeper analysis. For example, the Literature Review Agent can provide summarized findings to the Research Gap Agent, which then identifies unexplored research opportunities. Similarly, the Trend Analysis Agent can share emerging topic insights with the Citation & Recommendation Agent to suggest relevant papers and references.



Technology Used

- **Langflow platform** -Used for designing and orchestrating multi-agent workflows visually for the Research Agent system.

- **IBM Grainte model-** For example, ibm-granite-3-2-8b is used for natural language understanding, reasoning, summarization, and intelligent research analysis.

- **IBM Cloud** – Provides the infrastructure to build, deploy, and scale the Research Agent efficiently in a secure and reliable environment.

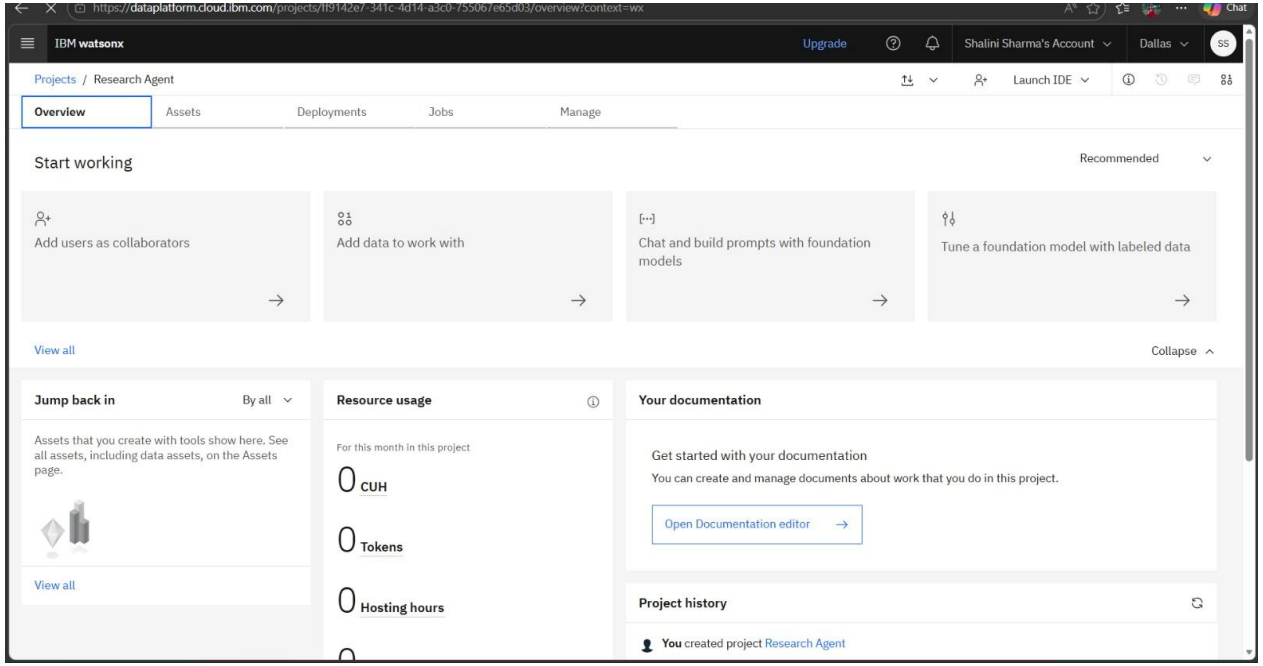
- **RAG–** RAG to **fetch real data first, then generate intelligent answers**, making the Nutrition Agent more trustworthy and effective.

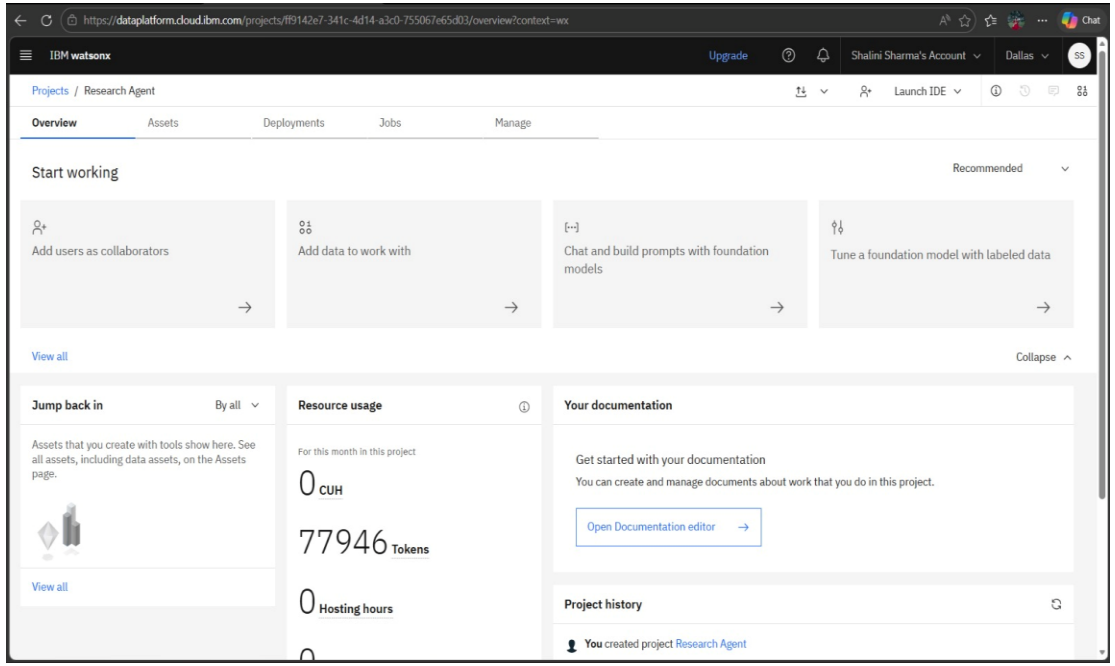
- **IBM Watsonx.ai -** Used for model deployment, embedding generation, AI governance, and integration of IBM Granite models within the Research Agent workflow.

- **Vector Database** (FAISS/Chroma) – Used for storing embeddings and enabling semantic search for efficient RAG-based research paper retrieval and contextual information access.

Token Usage

77,946 token are used





Novelty and Uniqueness

The proposed Research Agent stands out by combining Agentic AI, RAG, semantic search, and intelligent research automation into a single advanced system.

Multi-Agent Intelligence – Instead of relying on a single chatbot, multiple specialized agents collaborate to perform literature review, research gap identification, trend analysis, citation generation, and intelligent research assistance.

RAG-Based Accuracy – Integrates Retrieval-Augmented Generation (RAG) with real-time research data retrieval, reducing hallucinations and improving the accuracy and reliability of generated research insights.

Semantic Search Capability – Uses embeddings and vector databases to perform context-aware semantic search instead of simple keyword matching, enabling more relevant research retrieval.

Automated Literature Review – Automatically summarizes and compares multiple research papers, saving significant time and effort for researchers and students.

Research Gap Identification – Detects unexplored areas, limitations, and future research opportunities from existing academic studies.

Trend Analysis & Insights – Identifies emerging technologies, trending topics, research patterns, and frequently discussed methodologies across domains.

Citation & Recommendation Support – Generates structured citations and recommends relevant research papers, journals, and academic resources.

Multi-Source Data Processing – Supports PDFs, research papers, URLs, academic repositories, and text documents for comprehensive research analysis.

Scalable & Enterprise-Ready – Built using IBM watsonx.ai and IBM Granite models, ensuring scalability, reliability, AI governance, and secure deployment.

Git Hub Link

· Please create public github repository (Enable Add README button while creating repository) with format

Please test and Paste the working Github URL – https://github.com/hiremathchinmayi77-hash/Research-Agent

Make sure you have Uploaded following three files into your github repository

1) app.json file , 2) yourproblemstatement.pdf file 3) your projectpresntation.pptx file 4)any other files

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Future Scope

The future scope of the AI-powered Research Agent is extensive, as advancements in Agentic AI, Large Language Models, and Retrieval-Augmented Generation (RAG) can further enhance its capabilities. The system can evolve into a fully autonomous research assistant capable of supporting complex academic, industrial, and scientific research activities.

Advanced Multi-Agent Collaboration – Future versions can include more specialized agents for plagiarism detection, hypothesis generation, experiment analysis, patent research, and automated report writing.

Multimodal Research Support – The system can be enhanced to analyze not only text documents but also images, graphs, tables, audio lectures, research videos, and handwritten notes.

Real-Time Research Monitoring – Integration with live academic repositories and journals can allow the agent to continuously track newly published papers, emerging technologies, and trending research topics.

AI-Assisted Paper Writing – The Research Agent can assist users in drafting complete research papers, abstracts, methodologies, and conclusions with intelligent formatting and suggestions.

Automated Citation & Reference Management – Future improvements can support advanced citation validation, duplicate reference detection, and integration with tools like Zotero and Mendeley.

Cross-Domain Knowledge Discovery – The system can connect research insights from multiple domains such as healthcare, AI, finance, and engineering to generate interdisciplinary innovations.

Personalized Research Recommendations – AI models can learn user preferences and recommend research papers, journals, conferences, and project ideas based on individual interests and academic history.

Thank You!

Thank you for your time and interest.